A viscous rung condition for layered forced blowup, and a forcing-gap formulation of Navier–Stokes regularity

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Abstract

Alpöge and Buckmaster have constructed finite-time blowup with smooth forcing for the incompressible porous medium equation, the 2D Boussinesq system and 3D incompressible Euler, following the multiscale program of Córdoba and Martínez-Zoroa. Their engine is an exact affine-background wave whose amplitudes obey a two-by-two linear system, with a controlled background rotation that returns each layer's vorticity to zero before the next layer grows. We add dissipation $\nu(-\Delta)^\alpha$ to that amplitude law and read off the condition for a layer at frequency λ_q to grow, $\Gamma_q \gtrsim \nu\lambda_q^{2\alpha}$. Under the published frequency schedule $\lambda_q = \lambda_{q-1}^Q$, $Q \geq 201$, with growth scale $\sigma_q \sim \lambda_q^{1/16}$, the ladder as written locks at every rung only if $\alpha < 1/(32Q)$; for $\alpha = 1$ it misses the second rung by roughly 240 orders of magnitude at $\nu = 10^{-3}$. This is a statement about the ansatz class and schedule, not a theorem about the Navier–Stokes equations. We then define a scale-covariant force budget $J(f)$ for a blowup pair (u_0, f) and observe that a positive lower bound $J(f) \geq \delta > 0$ over all blowup pairs implies global regularity for the unforced equations, while the forced constructions supply an upper bound on δ. Two registered numerical probes are reported: the amplitude law with dissipation, and a localized vorticity layer on a stagnation strain in a 64^3 Navier–Stokes box, where the sign and ordering of the rung condition hold and the absolute growth rate falls 0.2–0.25 below the naive prediction. Nothing here proves or disproves any of the four Clay statements.

1 Setting

Fefferman's official problem statement has four options. (A) and (B) ask for global smooth solutions from every smooth, divergence-free, finite-energy datum on $\mathbb{R}^3$ or $\mathbb{T}^3$ with zero external force. (C) and (D) ask for one smooth datum and one smooth, bounded-energy force $f(x,t)$ whose solution breaks down in finite time. The force in (C) and (D) is part of the official wording. Alpöge and Buckmaster [1, 2, 3] establish the forced statements for Boussinesq and 3D Euler in the smooth-forcing class; a forced Navier–Stokes result has been announced by OpenAI but not released at the time of writing. The question this note is about is the unforced one.

2 The amplitude law and where the force enters

We follow [1, Lemma 3.1]. Suppose the fields already constructed are affine on a region, $u_{\text{old}} = D(t)x$ with $\operatorname{tr} D = 0$ and $\theta_{\text{old}} = G(t) \cdot x$. Add a wave with phase $s = \lambda\,\zeta(t) \cdot x$, temperature

$\vartheta = \Theta(t)F(s)$ and vorticity $\varpi = \Omega(t)F'(s)$ for a smooth odd 2π-periodic profile F. The scalar and vorticity residual increments vanish exactly if and only if

$$\dot{\zeta} = -D^T\zeta, \qquad \dot{\Theta} = -\frac{J\zeta \cdot G}{\lambda|\zeta|^2}\,\Omega, \qquad \dot{\Omega} = \lambda\,\zeta_1\Theta. \tag{1}$$

For the frozen example $D = 0$, $G = -Ae_2$, $\zeta = r\,e(\varphi)$ the eigenvalues are $\pm\sqrt{A\sin\varphi}$. The growth rate of layer q is therefore set by the gradient left behind by all earlier layers, $\sigma_q^2 = |G_{<q+1}|$, with rate $\Gamma_q = \sigma_{q-1}\sin s_q$ [1, (3.10)]. The 3D Euler construction [2, (3.4)] has the same structure with circulation and azimuthal vorticity as the two fields.

The force enters in four places in [1]: the base rotation $Du^b(0,t) = \dot{\alpha}(t)J$ [1, (9.1)], the smooth activation of each layer, the nonoscillatory phase averages of the localization errors [1, (6.6),(6.11)], and the compact recovery of the force through an inverse curl [1, Lemma 6.2]. The last three are bookkeeping. The first carries the physics: a rigid rotation with time-varying rate is not a solution of the unforced equations, since $\partial_t u = \ddot{\alpha}Jx$ is not a gradient, and it is precisely the control $\dot{\alpha} = \mathcal{F}_q - \dot{Z}_q/\zeta_{q,2}$ [1, (3.12)] that steers each layer's vorticity amplitude back to zero so the next layer sees no shear. Unforced, the equations must supply this steering from the pressure alone.

3 The viscous rung condition

Proposition 1 (Rung condition for the affine-wave class)**.** *Add dissipation $\nu(-\Delta)^\alpha$, $\nu > 0$, $\alpha \in (0, 1]$, to both wave equations in the frozen growth configuration of* (1)*. The amplitude system becomes*

$$\frac{d}{dt}\begin{pmatrix}\Theta\\ \Omega\end{pmatrix} = \begin{pmatrix}-\kappa & A\sin\varphi/(\lambda r)\\ \lambda r\sin\varphi & -\kappa\end{pmatrix}\begin{pmatrix}\Theta\\ \Omega\end{pmatrix}, \qquad \kappa = \nu(\lambda r)^{2\alpha},$$

whose growing eigenvalue is $\mu = \sqrt{A}\,\sin\varphi - \nu(\lambda r)^{2\alpha}$. The layer amplifies if and only if

$$\sqrt{A}\sin\varphi > \nu(\lambda r)^{2\alpha}. \tag{2}$$

If only the scalar field is dissipation-free (Boussinesq with inviscid temperature), μ solves $\mu^2 + \kappa\mu - A\sin^2\varphi = 0$ and is positive for every κ, with $\mu \approx A\sin^2\varphi/\kappa$ when κ dominates.

Proof. Direct computation of the characteristic polynomial in each case. □

Proposition 2 (The published schedule under dissipation)**.** *Take the schedule of [1, (3.7),(3.18),(3.52)]: $\lambda_q = \lambda_{q-1}^Q$ with $Q \geq 201$, growth scale $c_\sigma\lambda_q^{1/8} < \sigma_q^2 < C_\sigma\lambda_q^{1/8}$, rate $\Gamma_q = \sigma_{q-1}\sin s_q \leq \sigma_{q-1}$. With both fields damped,* (2) *at rung q reads $\lambda_{q-1}^{1/16} \gtrsim \nu\lambda_{q-1}^{2\alpha Q}$ up to fixed constants. Hence for every fixed $\nu > 0$ the ladder locks at all sufficiently large rungs only if $\alpha < 1/(32Q)$, and for $\alpha \geq 1/(32Q)$ it fails at every rung beyond a finite index depending on ν.*

Proof. Insert the schedule into (2): the damping at rung q acts on the new frequency $\lambda_q = \lambda_{q-1}^Q$ while the growth uses $\sigma_{q-1} \sim \lambda_{q-1}^{1/16}$; compare exponents of λ_{q-1}. □

Remark 1. Proposition 2 concerns the construction as published. A different amplification mechanism, or a schedule in which the earlier layers leave behind a gradient growing like $\nu\lambda_q^2$, is not covered. What the proposition says is that reaching $\alpha = 1$ within this class requires the background strain to grow like the square of the new frequency at every rung, which is the Burgers balance $s = \nu\lambda^2$: the core scale of a strained vortex is $\delta^2 = \nu/s$. Unforced Navier–Stokes sits at the margin of (2) on every coherent core, with no exponential gain to spend, while the scheme needs logarithmic gains that grow by a factor of at least 100 per rung [1, (3.52a)]. This is consistent with the only viscous case mentioned in [2], hypodissipative Navier–Stokes with a sufficiently small fractional exponent [5].

4 A forcing-gap formulation

Definition 1 (Force budget)**.** *For a smooth blowup pair (u_0, f) in the sense of Fefferman's (C)/(D), with breakdown time T_*, define*

$$J(f) = \int_0^{T_*} \|\nabla \times f(t)\|_{L^\infty}\, dt \qquad \text{or} \qquad J'(f) = \sup_{t<T_*} \frac{\int \omega \cdot (\nabla f)\, \omega\, dx}{\int |\omega|^2\, dx} \cdot (T_* - t).$$

Both are invariant under the Navier–Stokes scaling $u \mapsto \lambda u(\lambda x, \lambda^2 t)$, $f \mapsto \lambda^3 f(\lambda x, \lambda^2 t)$, which is why they are the candidates; a non-covariant budget cannot decide anything, for the reason recorded in [6, Lemma 4].

Lemma 1 (Gap implies regularity)**.** *If there is $\delta > 0$ with $J(f) \geq \delta$ for every smooth blowup pair, then statement (A) holds.*

Proof. A breakdown with $f \equiv 0$ would be a blowup pair with $J = 0 < \delta$. □

The forced constructions give the upper bound $\delta \leq J(f_{\mathrm{AB}})$ once the force is written explicitly. Proving the lower bound is the whole difficulty, relocated: on a Burgers-slaved core one expects a growth rate at most $|S| - \nu\lambda^2$ with $|S|$ bounded by the core's own circulation and strain, so a super-exponential ladder must be fed from outside the core, and in unforced flow "outside the core" means other cores. We do not claim this lemma.

5 Registered numerical probes

E1: the amplitude law with dissipation. Integrate the frozen growth stage of (1) with both fields damped, in log space, along the published schedule ($Q = 201$, $\lambda_1 = 4$, $A_q = \lambda_q^{1/8}$, $\sin s = 0.05$, $\nu = 10^{-3}$). Figure 1 shows $\log_{10}$ of growth rate over damping at rungs 1–4. The prediction registered before the run was that Navier–Stokes-type damping kills rung 2 for every $\alpha \geq 1/32$; it held, and the threshold is in fact $1/(32Q)$.

E2: a layer on a stagnation strain in the Navier–Stokes box. Pseudo-spectral vorticity solver, 64^3, 2/3 dealiasing, RK2, Taylor–Green background $u = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$ whose origin is a stagnation point with strain $\mathrm{diag}(1, -1, 0)$. A localized layer $\varepsilon\, e^{-r^2/2\sigma^2} \cos(\lambda y)\, e_x$, $\varepsilon = 10^{-3}$, $\sigma = 0.5$, is added to the vorticity; a control run without the layer is subtracted; the band enstrophy of the difference around $(0, \pm\lambda, 0)$ gives the amplitude growth rate. Registered prediction: rate $\leq S_{\mathrm{env}} - \nu\lambda^2$ within 10% and negative for $\lambda \geq 16$ at $\nu = 4 \cdot 10^{-3}$. Outcome (Figure 2): sign and ordering hold, with the sign change near $\lambda \approx 12$ at $\nu = 4 \cdot 10^{-3}$ against a predicted 14.5 and no change through $\lambda = 16$ at $\nu = 10^{-3}$; the absolute level fails the tolerance, sitting 0.2–0.25 below the prediction at every λ and ν in the first half-window and lower in the second as the strain compresses the layer. On the real equations the layer receives less than the stagnation-point strain, so the rung is at least as hard as Proposition 1 says.

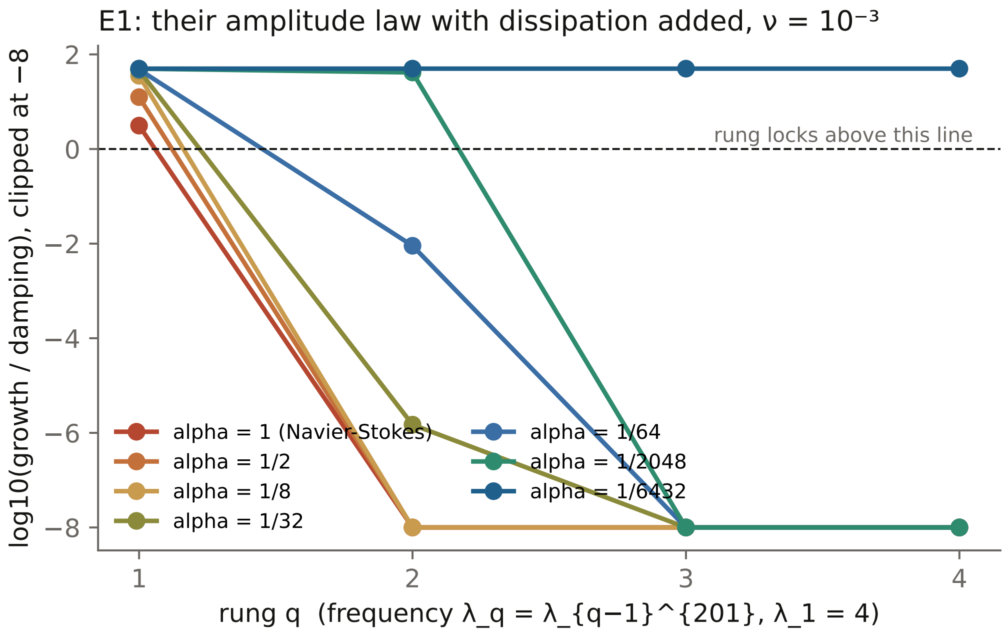

Figure 1: E1. Each curve is one dissipation exponent α; the ladder locks while the curve stays above zero. Only $\alpha = 1/6432 = 1/(32Q)$ locks every rung.

6 What is and is not claimed

Propositions 1 and 2 are elementary and concern the affine-wave ansatz and the published schedule. The forcing gap is a definition and a one-line lemma; the lower bound is open and is the unforced problem itself. The numerics are at $\nu \geq 10^{-3}$ on a periodic box and do not extrapolate to the inviscid limit. None of this proves or disproves any of Fefferman's four statements. The closed component lemmas, the open gap in four equivalent forms, and the rest of the July 2026 program are described at [6].

Acknowledgements

The July 2026 program and this note were carried out with two AI systems working adversarially under the author's direction, with every numerical claim registered before the run and every retraction kept on record.

References

[1] L. Alpöge, T. Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, preprint, September 2026, https://cims.nyu.edu/~tristanb/boussinesq.pdf.

[2] L. Alpöge, T. Buckmaster, *Blowup for the Euler equations with smooth forcing*, preprint, September 2026, https://cims.nyu.edu/~tristanb/euler.pdf.

[3] L. Alpöge, T. Buckmaster, M. P. Coiculescu, *Finite time blowup for the incompressible porous medium equation with uniformly smooth force*, preprint, 2026.

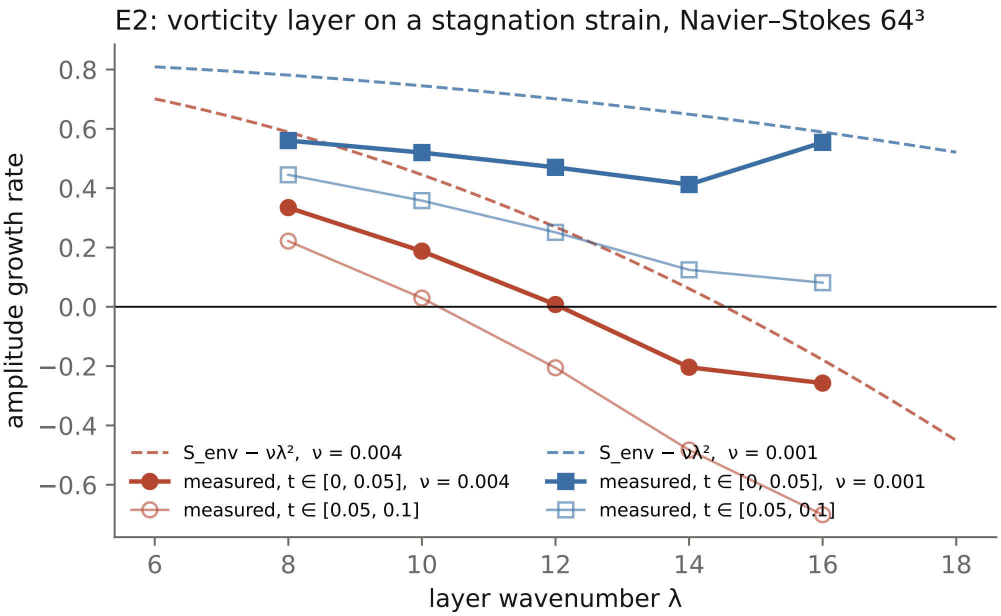

Figure 2: E2. Dashed: $S_{\mathrm{env}} - \nu\lambda^2$. Filled markers: measured rate on $t \in [0, 0.05]$; open markers: $t \in [0.05, 0.1]$.

[4] D. Córdoba, L. Martínez-Zoroa, *Finite time singularities for the forced incompressible porous medium equation*, and related works.

[5] D. Córdoba, L. Martínez-Zoroa, F. Zheng, *Blowup for hypodissipative Navier–Stokes with a sufficiently small fractional dissipation exponent and force*, as cited in [2].

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